

Power Electronics and Drives

Applications to Modern Ship Propulsion Systems

Electrically propelled ships gained popularity by the early 20th century, with the rapid development of submarines and medium-capacity container ships, mainly using dc motors [1]. Synchronous ac motors have since been employed for naval propulsion systems, but due to the restricted operation of the available power electronic devices at that time, these configurations were too expensive and unreliable, and they featured poor performance in terms of speed and torque control [2]. Improvements in power electronics devices and drive control schemes as well as the development of high-efficiency multi-phase induction and synchronous motors have pushed the advancement of electric ship propulsion systems and applications [3]–[7].

Depending on the power requirements and propulsion system arrangement, different electrical propulsion systems have been developed based on two-level and multilevel converters using series-connected and/or multiwinding electric machines [8]. In this context, several configurations have also been used that either integrate all electrical network components in only one power system or separate the mechanisms in several power systems, such as in warships, where propulsion systems are partitioned from weapon systems, due to the large amount of power required

for propulsion. New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large amounts of energy for pulsed-power weapons and large propulsion machines. Pulsed-power weapons have a significant energy demand, requiring large-capacity power systems and improved dynamic characteristics, compared to traditional and legacy ships. The conventional approach to meet the increasing energy demand is

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to integrate more generation sources and storage units into the ship power system [9], as, for example, in the Type 45 destroyer (U.K. Royal Navy) and the Zumwalt-class destroyer (U.S. Navy) [10]. Hybrid diesel/electric systems, where propulsion is provided by a combination of gas turbines, diesel engines, and electric motors, have been proposed as an intermediate step to improve the efficiency of vessels [11].

International Maritime Organization (IMO) ship pollution rules, in the International Convention on the Prevention of Pollution From Ships, which was ratified in 1973 and modified in 1978 [12], and the recently enforced IMO 2020 Global Sulfur Limit [13] have established limits for carbon dioxide, nitrogen oxide, and sulfur oxide emissions from ship exhaust gases. These stringent regulations have introduced new challenges for ship designers and owners, producing a significant impact on vessels' energy efficiency design indexes and energy efficiency operational indicators, leading to the development of both hybrid diesel/electric and full-electric propulsion systems [14].

A review of power system topologies, electric machines, and power converter topologies is presented in this article. Examples of ships and vessels that use the different configurations are given. Recent advances and future trends are also discussed.

Ship Propulsion System Classification

To specify a propulsion and transmission system configuration, the operating profile of a ship must be taken into account, in addition to the amount of power, to consider the vessel type and function. The operational profile defines the required power demand for thrust in different modes. For example, commercial and passenger vessels typically operate at a constant speed that is close to their design rating, requiring large power drives with high-torque, low-speed, and low-noise machines and converters. Cargo, oil, and gas vessels require a relatively fast speed and high fuel efficiency. On the other hand, icebreakers need significant torque, travel at low speeds, and

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demand high performance to deal with constant load variations. In tugboats, the main propulsion must be oversized to meet the bollard-pull requirements.

In the case of naval combat vessels, low speeds and reduced power demands are necessary for patrols and transit, but high speeds and a greater power output are required for tactical operative purposes. In warships, a combination of performance, redundancy (survivability), and flexibility are necessary. In conventional submarines, the power source is inherently electrical, such as battery storage and fuel cells. Submarines and warships also need to generate very little acoustic noise [15]. A general classification of different propulsion system configurations, according to transmission systems and energy sources, is presented in Figure 1. Propulsion systems can be primarily classified as geared, in which an amplifying or reductive gearbox is used to connect the gas turbine, diesel engine, or electric motor

to the main propeller, and as gearless, in which the machine is directly connected to the propeller.

According to the energy source, propulsion systems can be classified as thermal (i.e., diesel engines and gas turbines), electric, and hybrid. Thermal and hybrid configurations commonly incorporate a gearbox to resolve the differences between the prime mover and the propeller speed and achieve maximum efficiency, which is typically below 200 r/min for propellers, between 700 and 1,000 r/min for high-speed diesel engines, and roughly 3,000 r/min for gas turbines. A gearbox is, therefore, essential to “match” the speeds and provide a mechanical coupling in combined thermal and hybrid configurations. On the other hand, electric propulsion systems are often gearless because electrical machines can be designed for low-speed operation optimized to suit the requirements of the propeller. Hence, the development of full-electric propulsion

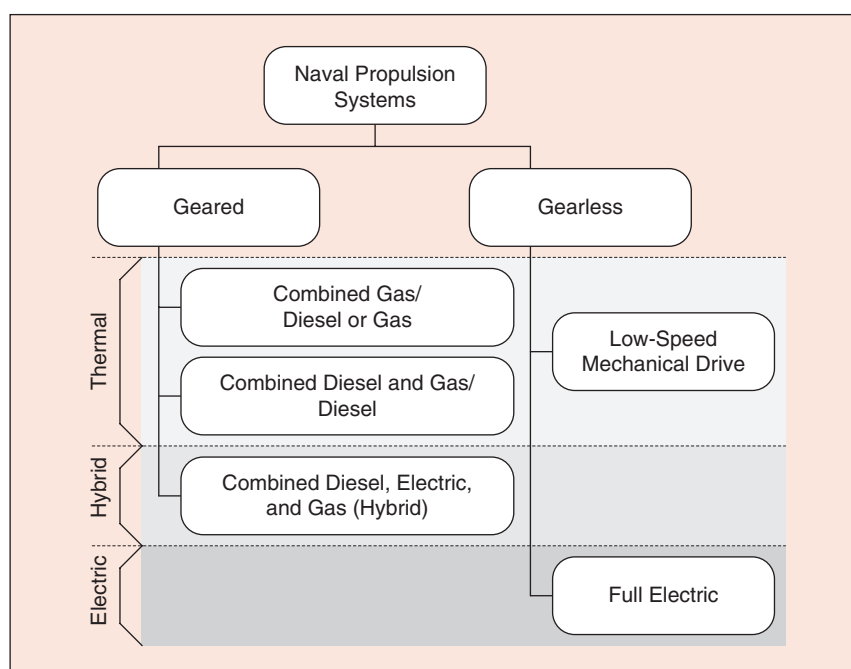


FIGURE 1 – The classification of ship propulsion systems, according to transmission system and energy sources.

The concept of the advanced hybrid drive system has been proposed for fuel savings at low ship speeds.

systems would not only maximize efficiency and minimize emissions but also reduce the required space.

Thermal Combined Propulsion Systems

Thermal propulsion systems are commonly based on a combination of different thermal cycle machines (i.e., diesel engines and gas turbines) with different nominal speeds, which are selected to work alone or together via a synchronous clutching system, depending on the operating requirements of a vessel. The propulsion system includes a shafting arrangement and the propeller. Depending on the application and operational power requirements, the most suitable mechanism must be selected. For example, in naval combat ships, a diesel engine is used for patrols, and a gas turbine is employed for high-speed maneuvering. The main disadvantages of combined propulsion systems relate to the complexity of the gearing system and the large machinery space required

for the main and auxiliary equipment, as shown in Figure 2(a).

Hybrid Propulsion Systems

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors, as illustrated in Figure 2(b), to provide a wide range of propeller speeds. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. However, the main drawback stems from the oversized design of the main generators that is necessary to provide power to the electric machine for propulsion and the ship auxiliary services.

For the aforementioned reasons, the concept of the advanced hybrid drive system (AHDS) has been proposed for fuel savings at low ship speeds [16]. A classical hybrid propulsion system appears in Figure 3, where a gas turbine is used for high speeds and an electric machine is employed

for low speeds so as to avoid a reduction of the gas turbine's efficiency. In this configuration, the electric machine is fed by a bidirectional back-to-back power converter that can operate the machine either as a motor or a generator. This configuration enables a ship to operate entirely in electric drive mode, taking energy from the main busbar (the power take-in) so that the gas turbines can remain offline. If large gas turbines are installed, the electric machines can operate in generator mode, supplying power to the main busbar (the power takeoff) and supporting the power delivered by the main generators so that fewer generator sets are required in operation. The AHDS system offers several benefits over classical thermal-based combined propulsion systems, including efficient gas turbine operation at low ship speeds, additional power for sensor and weapons systems, and an efficient power dispatch for the main ship generators. The AHDS has been proposed as a new configuration for naval platforms, such as the U.S. Navy's Landing Helicopter Dock 8 amphibious assault ships [17], [18].

Full-Electric Propulsion Systems

Full-electric propulsion is a configuration in which the main prime mover is

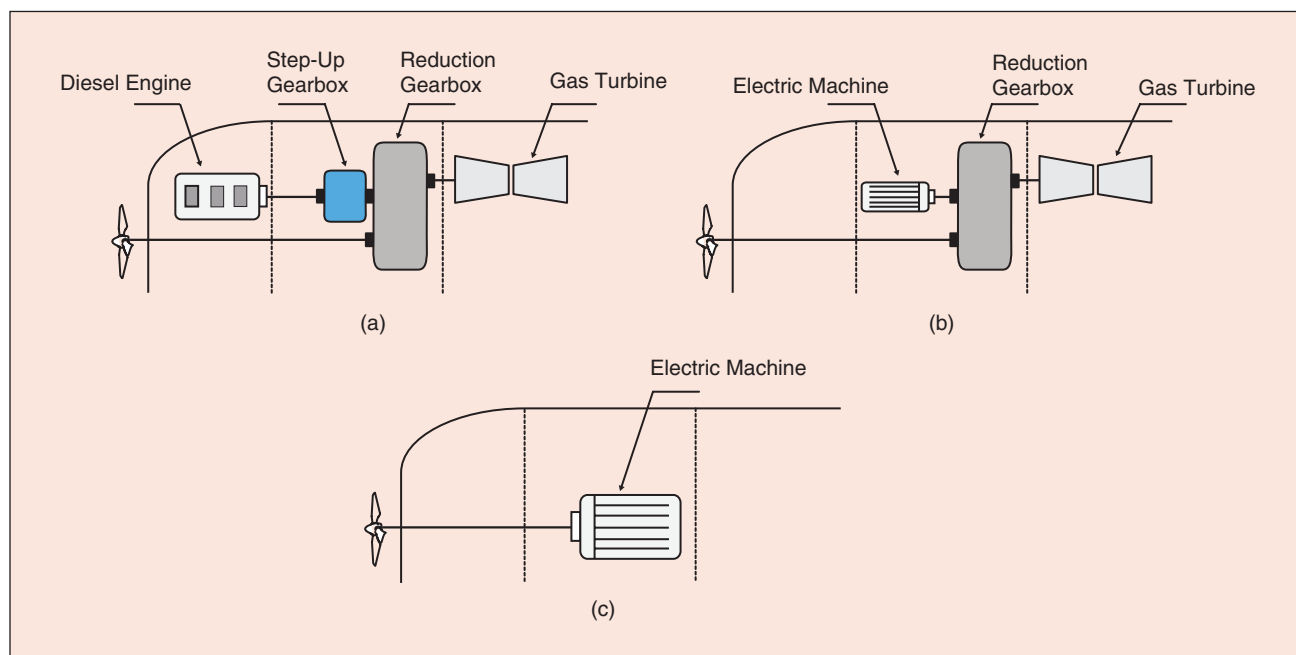


FIGURE 2 – Ship propulsion systems. (a) Thermal combined. (b) Gas electric. (c) Full electric.

an electric motor directly connected to the propulsion shaft, with or without a gearbox. Electric power is provided by generators operated by a gas turbine, a diesel engine, or a combination of both. The main advantages of full-electric propulsion include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller. Moreover, outer azimuthal pods also offer flexibility to allocate the generators elsewhere on a ship (they do not require dedicated machinery spaces), as shown in Figure 2(c), thus reducing the required mechanical area and providing extra watertight compartments, which improves a ship's stability conditioning (machinery spaces are flooded areas). According to its structure, electric propulsion systems can be classified as segregated or integrated, as explained in the following.

Segregated Power System

The segregated power system (SPS) is the natural evolution of electrically propelled vessels, resulting in two separate electrical systems, one dedicated to providing power for propulsion and the other reserved for

feeding auxiliary loads, as presented in Figure 4(a). As a consequence of this particular arrangement, the generator sets dedicated to each system do not have the ability to interact with other systems since they have their own voltage and frequency. In the SPS, nearly 90% of the installed power is dedicated to propulsion, making this configuration very attractive for applications where a high level of maneuverability is required, such as icebreakers and towing ships [19]. However, most of the time, the generators operate below the rated load, increasing emissions. The excess power is not available for the auxiliary loads, hence reducing the power grid flexibility and as well as the availability and redundancy of the propulsion system.

Integrated Power System

In this architecture, the total power required by the propulsion system and the service loads is supplied by several generators connected to the medium-voltage bus and then distributed to the low-voltage bus through transformers. The propulsion motors are fed from power converters, which are connected directly to the main busbar, and the service loads are fed from a low-voltage busbar, using step-down transformers, as described in Figure 4(b). The integrated power system (IPS) combines the generation and distribution system into a unified grid, where the generated power is shared between all loads. The main features of an integrated grid architecture include flexibility, availability,

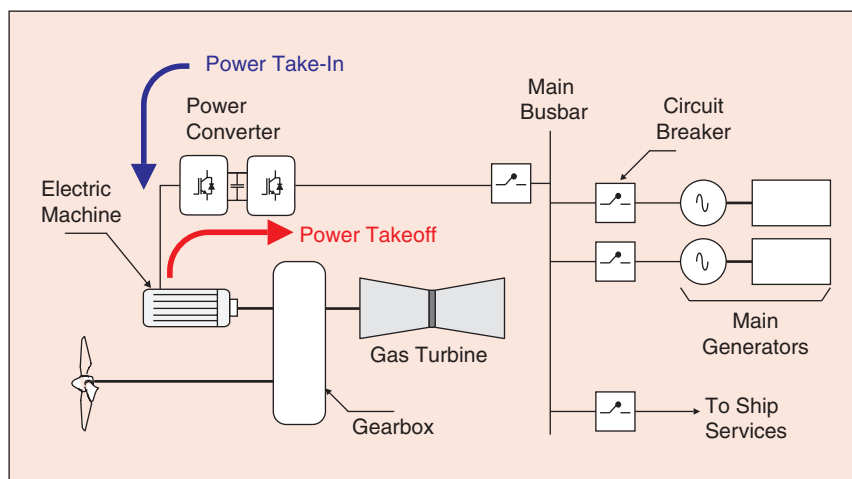


FIGURE 3 – AHDS gas and electric hybrid propulsion systems.

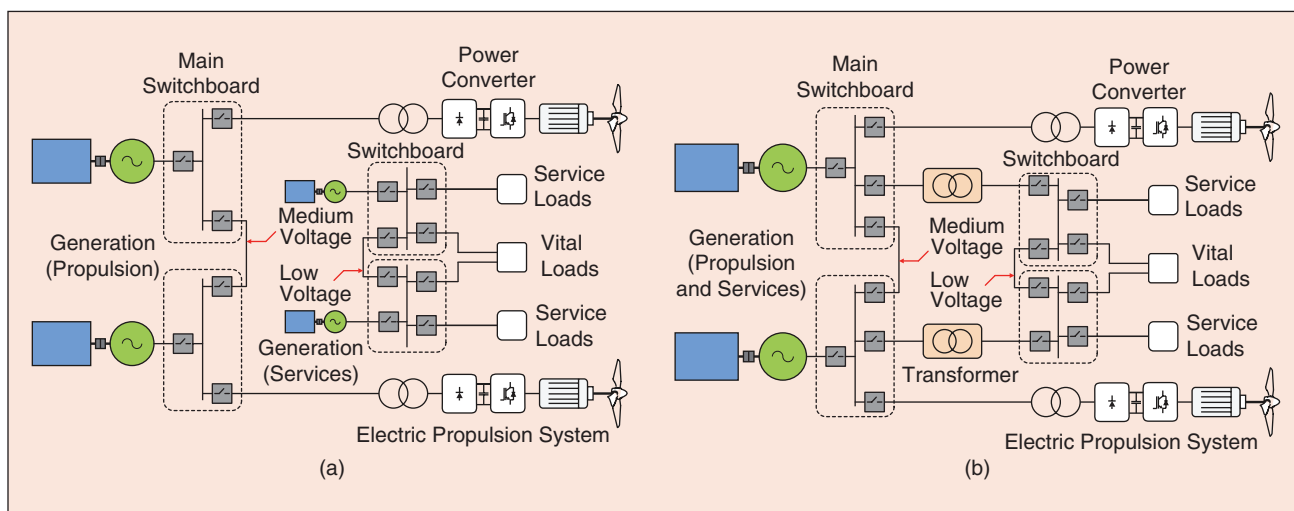


FIGURE 4 – Full-electric propulsion systems. (a) Segregated. (b) Integrated.

The development of permanent-magnet synchronous machines and high-efficiency induction machines has reduced the application range of dc machine drives.

and system redundancy. The IPS also offers the ability to manage the generated and delivered power, optimizing the number of running generator sets, which enhances fuel efficiency and reduces emissions.

Machines for Electric Propulsion Systems

Electric motors used in ship propulsion systems are typically high torque and medium to low speed. Among the main types of equipment used in this application are brushed dc machines, induction machines, synchronous machines, wound-rotor machines, and permanent-magnet machines. Recent research has investigated homopolar and high-temperature superconducting machines [14]. Synchronous reluctance machines (SynRMs) also appear to be an attractive alternative for marine electric propulsion applications.

Brushed dc Machines

Brushed dc machines have their stator field fixed to the stator winding, while the rotor field turns with the rotor. To make sure that torque is generated, the commutator circuit fixes the orientation of the rotor field while the armature winding rotates. Thus, both the stator and the armature fields are orthogonal to each other, ensuring maximum torque production.

Brushed dc motors have been in use for marine propulsion proposes for more than a century. Until the 1980s, the conventional (brushed) dc machine was the most common choice for high-demand industrial applications where high-performance speed and torque control was required, becoming the natural choice for naval propulsion systems. However, dc drive systems have been replaced by high-efficiency ac drives (vector control). Today, dc commutator motors are used in some specific naval vessels, where a high torque, low speed, and fast

control response are needed across a wide range of speeds, e.g., icebreakers, conventional submarines, tugboats, minesweepers, seagoing dredges, and oceanographic vessels.

The main advantages of brushed dc machines concern the simplicity of the torque and speed control scheme, which is very attractive choice for icebreakers, where rapid speed control and a high torque output leads to more effective and rapid forward/reverse operation to ram and break up ice formations. A typical icebreaker dc propulsion system dating to the late 1960s is presented in Figure 5, including a reconfigurable dc power system that is required to provide full-range speed control for normal shipping conditions and low-speed, high-torque control during icebreaking operations.

A more recent application of dc machines is the hybrid combined diesel-electric and gas propulsion system of the Type 23 frigates of the Royal Navy, dating to the late 1980s. The dc propulsion drive consists of two 3-MW GEC dc motors. The armature is fed from a six-pulse thyristor rectifier, and the field circuit is fed from a six-pulse antiparallel rectifier, as displayed in Figure 6. This arrangement enables regenerative breaking. However, because thyristor-based rectifiers are used, large filters are required, reducing the overall power density of dc drives compared to ac drives of similar power.

Currently, Siemens still manufactures a dc propulsion system, the SINAVY DC-Prop, for Type 209 conventional (battery-based) submarines, which are the latest of their class to be commissioned (in 2019). The propulsion system consists of two separate dc excitation motors arranged on a common shaft and connected either in parallel or in series, depending on the speed step, which is controlled via armature and field circuit dc choppers.

When operating with the armatures connected in parallel, auxiliary series windings are switched into the armature circuit to achieve the load distribution (these auxiliary windings are not used in a series connection). Four diesel generators provide the required battery charging capability.

The main disadvantage of dc machines concerns the periodic maintenance required by the commutator and brushes. High-power machines also need additional interpole windings to assist the commutation process, reducing the sparking in the commutator brushes. Another limitation relates to the maximum allowable operational voltage of up to 700 V, due to the insulation in the commutator. The development of permanent-magnet synchronous machines (PMSMs) and high-efficiency induction machines has reduced the application range of dc machine drives. However, current developments of dc power distribution systems in modern ships and the improvement of machine design techniques have revived efforts toward more efficient dc-based propulsion systems [3].

Induction Machines

Induction machines are very attractive for applications in ship propulsion because of their constructive simplicity, the ability to work under extreme-torque-demanding conditions, and almost maintenance-free operation. The main advantage of induction motors over dc brushed machines results from the fact that induction motors do not require a commutator circuit. Thus, the rotor field is generated by induction, becoming more compact and efficient and gaining a high power density. Moreover, these features enable the machine to operate through a wide range of load conditions. Depending on the construction of the rotor, induction machines are commonly classified into two groups: squirrel cage induction machines (SCIMs), where the rotor consists of bars shorted together by end rings, and wound-rotor induction machines (WRIMs), which have a three-phase winding in a short circuit allocated in a slotted rotor. The main difference between rotor

winding configurations concerns their output torque characteristics and power density.

The first serious attempt to use induction machines was made during the early 1920s on the USS *New Mexico*, the first ship with a turboelectric propulsion system, which consisted of nine Babcock & Wilcox boilers

feeding two turbogenerators to drive four 7,000-hp induction motors. This arrangement offered more power output and required far less space for the machinery, compared to a thermal-based propulsion system using steam turbines. However, the propulsion system needed a complex switchboard to adjust the induction machine rotation

speed and direction by changing the number of pole pairs and interchanging phases via various winding connection arrangements, making this application less successful than synchronous and dc machines. This was mostly due to the difficulty of developing an accurate shaft rotation speed, the complexity of the switchboards

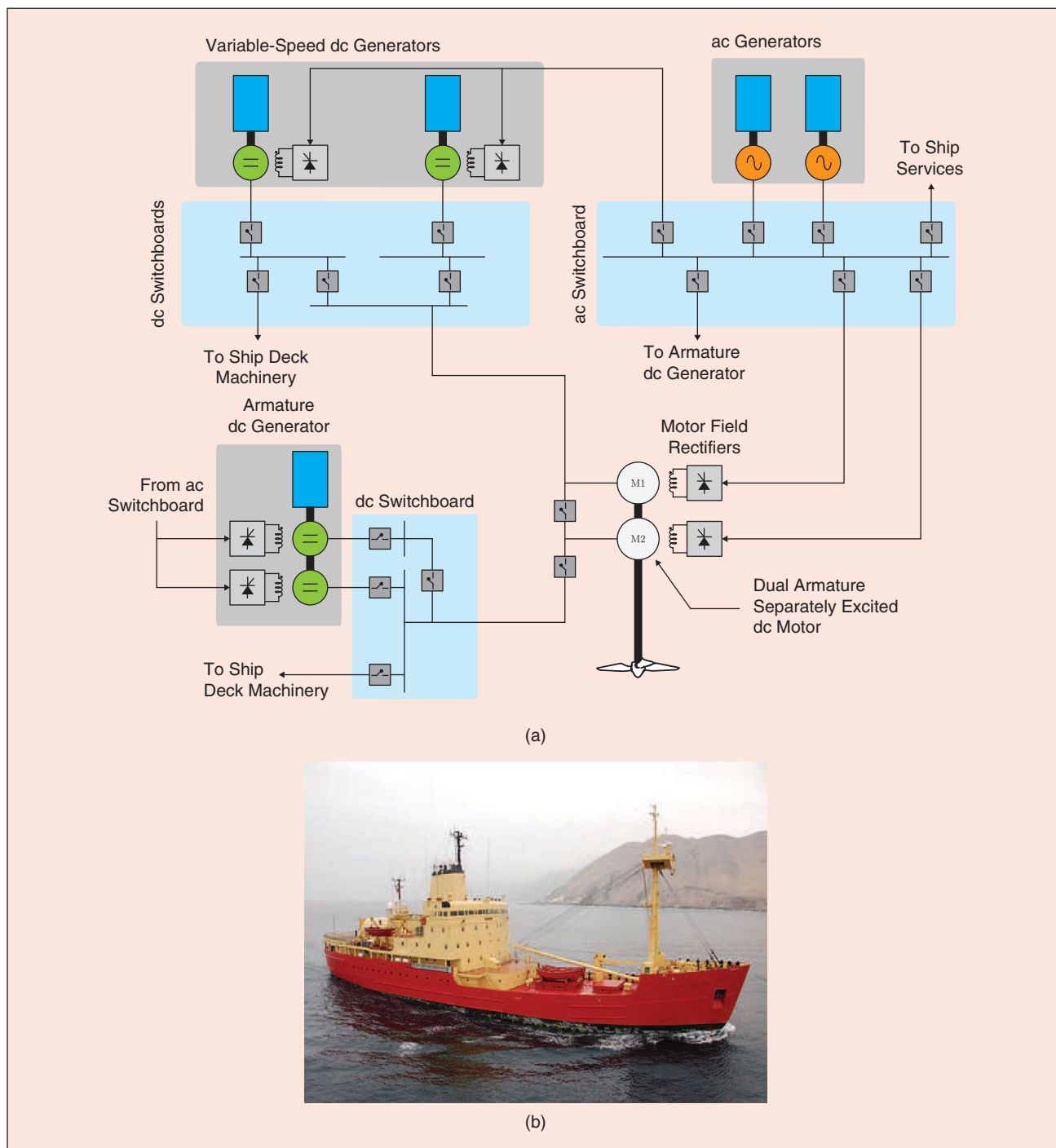


FIGURE 5 – (a) The Auxiliary Personnel (AP)-46 *Almirante Óscar Viel* dc propulsion system configuration (only one shaft is shown). (b) The AP-46 *Almirante Óscar Viel*.

used to control the rotation shaft speed, and a low efficiency due to a lack of torque control schemes. With the development of modern power electronics and vector control techniques, induction machine drives have become much more efficient in terms of their torque and speed control capability.

During recent years, Convertteam (a division of General Electric since 2011) has been among the leaders in induction motor development for marine applications, with the company's advanced induction motor (AIM). The AIM consists of a multiphase, shock-hardened, and high-torque-density induction machine that has an optimized electromagnetic

and cooling design. It has been specially developed for applications where a low speed and high torque are required, such as ship propulsion drives. The HMS *Queen Elizabeth*, pictured in Figure 7, is the first electrified aircraft carrier configured according to the IPS concept, consisting of two 35-MW gas-turbine ac generators, two 8.5-MW

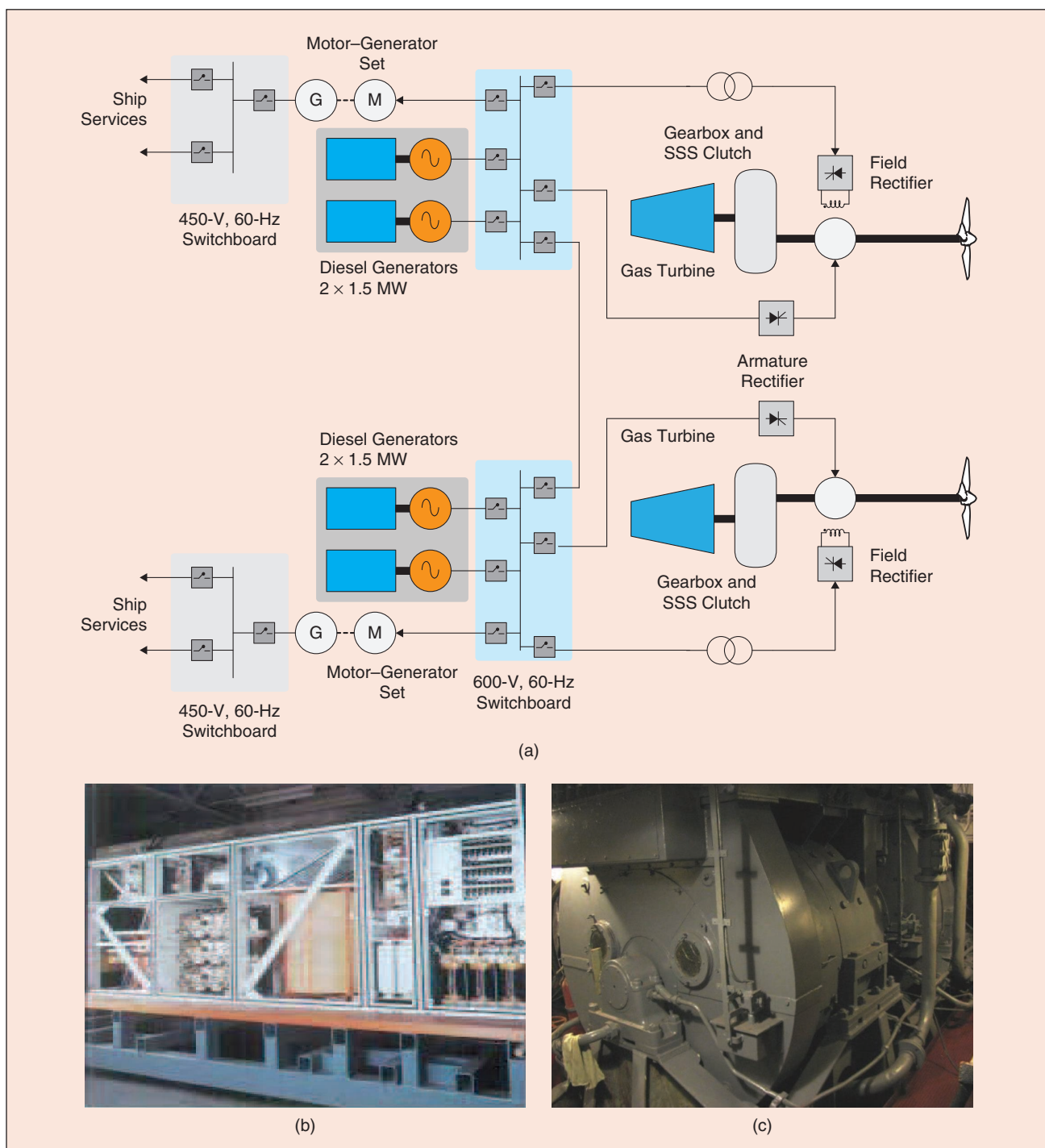


FIGURE 6 – (a) The Type 23 frigate hybrid propulsion system configuration, including (b) the armature and field converter and (c) the GEC dc motor. SSS: synchro self-shifting.

diesel generators, and four 20-MW multiphase AIMS in a two-shaft tandem configuration; each induction motor is fed by a Convertteam VDM25000 with a three-channel converter.

Synchronous Machines

The unique features of these machines, such as high efficiency, higher power

density, and simple speed control, compared to early developed dc drives, made the devices the ideal choice for replacing dc machines as a ship propulsion motor. Depending on how the rotor field is produced, synchronous machine rotors are classified either as wound-rotor or permanent-magnet machines. In early developments of

wound-rotor synchronous machines (WRSM), speed control was achieved by a variable-speed ac generator combined with the ability to change the number of poles on each winding. Later developments use load-commutated inverters to increase performance. The main disadvantages of WRSMs result from the need for slip rings and

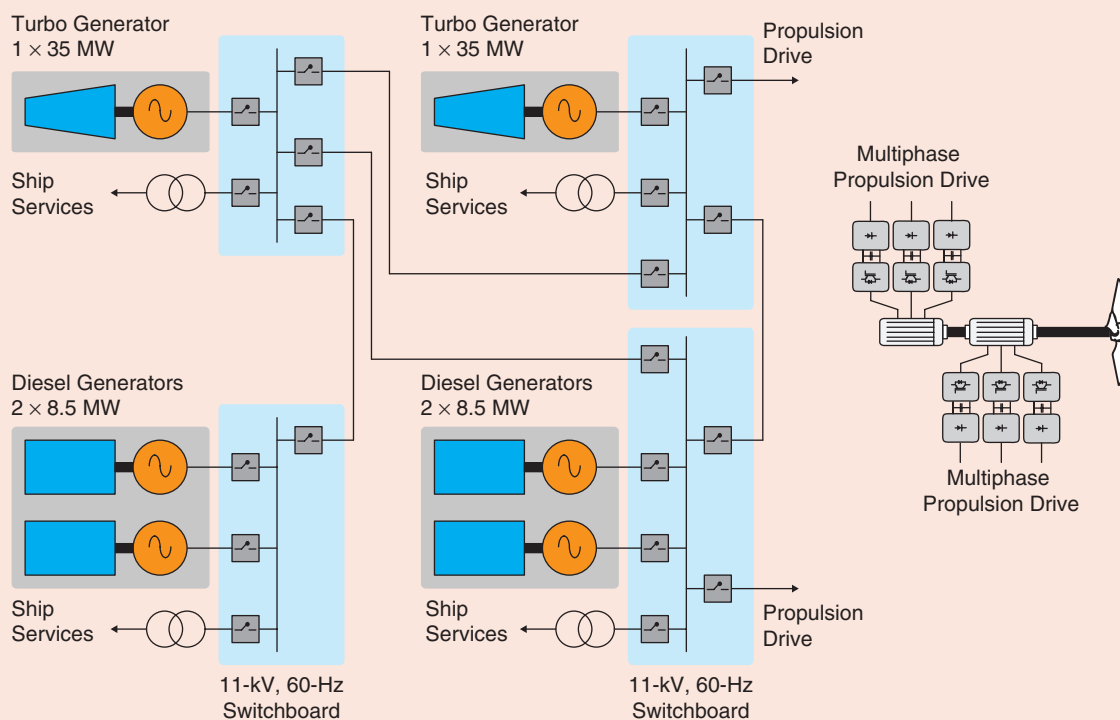


FIGURE 7 – (a) The HMS *Queen Elizabeth* integrated electric propulsion system based on two variable-speed Convertteam multichannel drives. (b) The HMS *Queen Elizabeth* Invincible-class carrier.

Induction machines are very attractive for applications in ship propulsion because of their constructive simplicity.

brushes to excite the rotor winding and the requirement for an external source to feed the field circuit.

PMSMs have the advantage of a higher power density compared to brushed dc, wound-rotor, and induction machines, due to the high flux remanence of permanent magnets. Moreover, PMSM efficiency reaches values of nearly 98%, while the efficiency of induction machines is limited to 94–96% at full loads. At partial loads, the PMSM shows its advantage compared to induction machines via a lack of losses resulting from induced rotor currents. The main disadvantages of the PMSM are related to the high cost of the magnets, the magnets' sensitivity to high temperatures, and partial demagnetization.

The SS *Canberra* was the first British cruise ship with an ac electric propulsion system, consisting of two WRSMs. The stator was fed by two ac turbogenerators via a winding pole-changing device. Four auxiliary turbogenerators drove an ac generator to power ship services and a tandem exciter for the propulsion generators, as shown in Figure 8. Other applications, especially in the field of PMSM, are in conventional (battery-based) and air-independent (fuel cell-based) submarines, such as the Scorpene and Type 212 classes, both propelled by a Siemens Permasyn 1.7-MW, high-density permanent-magnet synchronous motor.

Open-End Multiwinding Induction Machine

The open-end winding induction machine is composed of several three-phase windings connected to a power converter in each terminal [20], as in Figure 9(a), for a dual-star, open-end winding induction machine. Open-end winding induction machines offer several advantages for high-demand traction applications, such as independent phase current control, a greater number

of available active voltage vectors, and a superior degree of redundancy [21], [22]. This configuration enables operating the propulsion machine with one or both independent windings for different load requirements and in the event of a converter or machine winding fault [20], [24]. Under a winding fault condition, the three-phase winding may remain operational, modifying the current control scheme. Therefore, there is no need to isolate the faulty winding group while operating [25], [26]. All these advantages make multiple-phase winding machines a very attractive alternative in electrified ship propulsion systems [27], [28].

Doubly Fed Induction Machine

The doubly fed induction machine (DFIM) is based on a WRIM, with the rotor windings brought out through slip rings. This type of induction machine is widely used as wind generators for small- and medium-power applications [29], [30] as well as for traction [31]. For marine applications, the DFIM is connected to the main busbar via two power converters, which supply stator and rotor windings, respectively, as described in Figure 9(b). DFIMs are similar to SCIMs in terms of their torque capability and power density. However, the speed range of the DFIM can be extended because the stator and the rotor windings can interact with different voltage frequencies to control the induced back electromotive forces. Moreover, active power can be supplied by both the stator and the rotor windings, depending on operating conditions and regardless of speed. However, controlling both the stator and the rotor winding converted power flow adds extra complexity, and it does not substantially contribute to the propulsion system redundancy. Type 26 frigates currently under construction are

projected to be outfitted with a combined diesel-electric or gas-hybrid propulsion system with two 4-MW General Electric doubly fed induction motors.

SynRMs

SynRMs are based only on the reluctance torque due to the nonisotropic geometry of the rotor, as illustrated in Figure 9(c). Recently, the SynRM has gained attracted interest in the field of variable-speed industrial drives as a result of its higher torque, compared to induction machines with the same frame and power rating [32]–[34]. This leads to a higher power density and a lower rotor temperature, making the SynRM a very interesting alternative for marine power applications. In Table 1, a summary of the different electric propulsion machine drives for marine applications is presented, showing advantages, disadvantages, and application examples.

Power Converters

In electric ship propulsion systems, variable-frequency drives (VFDs) enable efficient and flexible operation according to a vessel's profile. However, the electronic power converters required to implement VFDs unavoidably introduce voltage and current distortion to a ship's power grid because of their switching operation, potentially causing additional losses, electromagnetic interference with sensitive equipment, torque pulsations, and grid resonances. Therefore, ship classification societies have set increasingly strict regulations for voltage harmonic distortion, where most applicable power quality stipulations, such as IEEE Standard 519, Military Standard 1399, and International Electrotechnical Commission Standard 61000-3-2, limit the voltage total harmonic distortion to less than 5%, with maximum individual harmonics lower than 3% [35]–[37]. The rectifiers installed to provide the dc link voltage for voltage source inverters (VSIs) in VFDs need to comply with harmonic distortion limits and tolerate medium-voltage

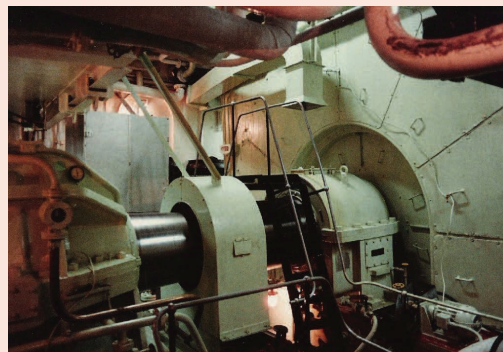
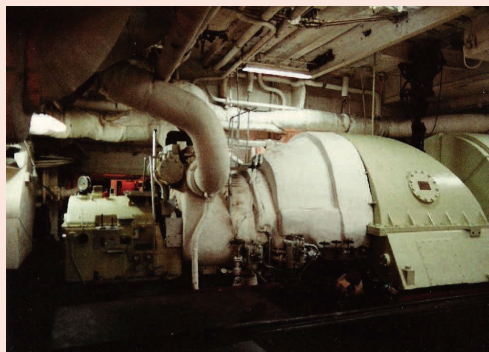
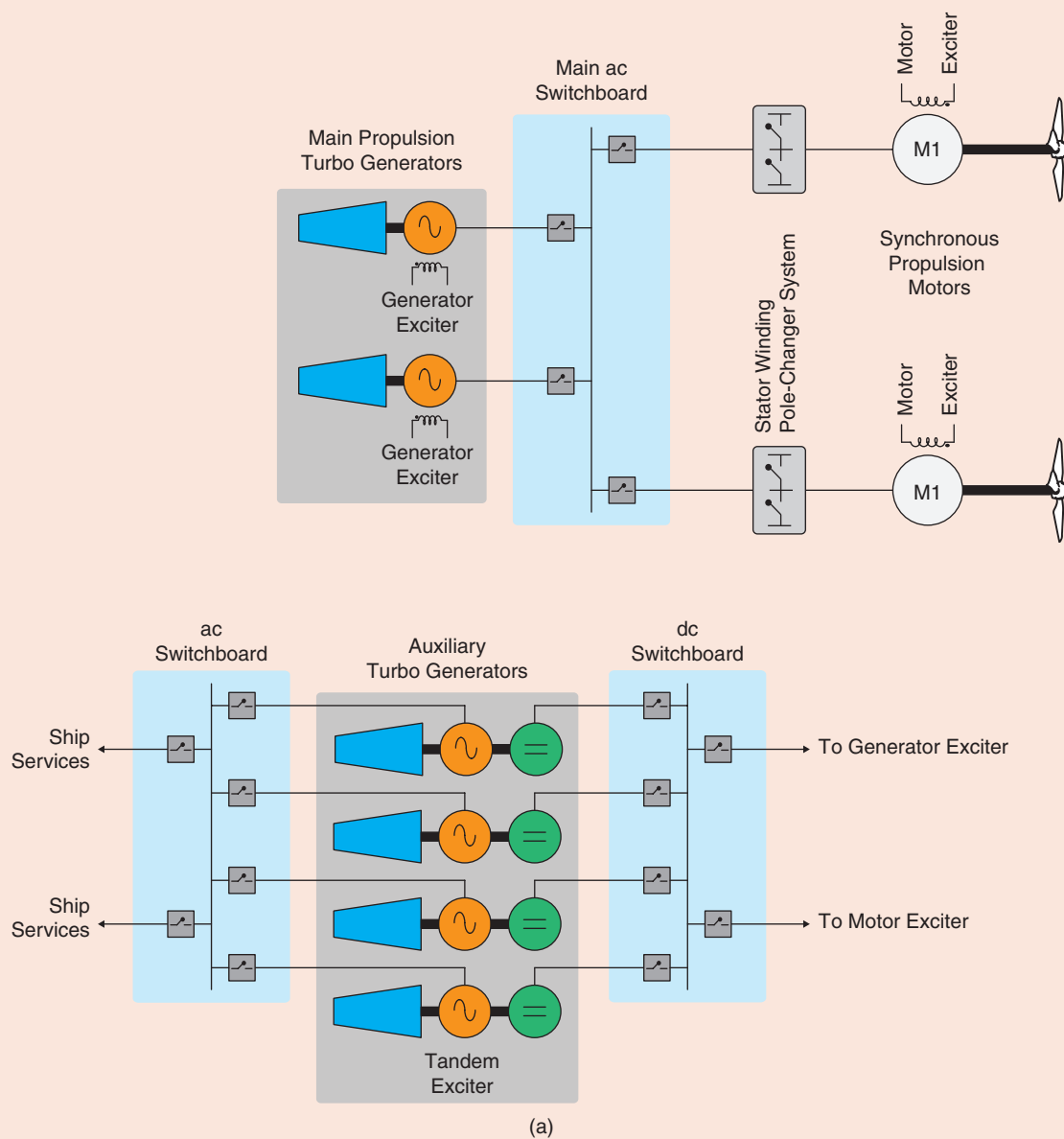


FIGURE 8 – (a) The SS *Canberra* ac propulsion system configuration, including (b) the main propulsion turbogenerator and (c) the main synchronous motors.

operation, as required by most modern electric ship propulsion systems in the megawatt range [38], [39]. The use of commercial off-the-shelf power converters has clear economic advantages for civilian and military applications in ship propulsion drives due to economies of scale and the availability of spare components for maintenance.

Rectifiers

Multipulse rectifiers are a mature technology that addresses two of the main issues associated with medium-voltage drive systems: harmonic distortion and the limited voltage rating of semiconductors. These rectifiers are composed of several six-pulse bridge rectifiers fed by phase-shifting transformers to achieve the cancellation of

low-order harmonics. Because the outputs of individual six-pulse rectifiers can be connected in a series, higher-voltage operation is achieved without requiring a full-voltage rating for individual diodes and silicon-controlled rectifiers (SCRs). A 12-pulse rectifier can mitigate more than 95% of the fifth and seventh harmonics [36], thus greatly reducing the voltage distortion, compared to a six-pulse rectifier. However, for the direct fulfillment of harmonic distortion limits without using additional filters, an 18-pulse rectifier might be required. In active front-end (AFE) rectifiers, the diodes and SCRs in the grid interface are replaced by integrated-gate bipolar transistor (IGBT) or integrated gate-commutated thyristor (IGCT) switches, which are capable of forced

commutation. This kind of rectifier facilitates input currents that are close to sinusoidal, a unity power factor, and a bidirectional power flow. However, a drawback of AFE drives results from a high-order (above the 50th) harmonic distortion that produces increased electromagnetic interference levels [35], [40].

Inverters

Inverters can be classified into three groups, according to the energy storage element installed in their dc links. VSIs have a capacitive dc link feeding a switched-voltage waveform to their loads. Current source inverters (CSIs) feature an inductive dc link. Hence, they feed switched currents to the load. Last, direct converters do not have a dc link and, therefore, do not need large energy storage elements, enabling them to reach very high power densities [41]. The converter topologies commonly employed in electric ship propulsion applications with ac drives are described in the following.

Two-Level VSIs

The two-level VSI (2L-VSI) is one of the most widely used power inverter topologies for industrial applications, due to the simplicity of its circuitry and modulation schemes. For electric motor drives, the most common configuration is an ac/dc/ac converter system implemented with a diode rectifier and a 2L-VSI, as in Figure 10(a). The diode rectifier can be replaced by an AFE rectifier if regenerative braking is required. There is a large number of providers of this technology, which is compatible with other industrial applications. Those companies include STADT [42], ENAG [43], General Electric [44], Siemens [45], Ingeteam [46], ABB [47], and The Switch (owned by Yaskawa) [48], all offering drives from 230 to 690 V. The application of this topology to medium-voltage drives is restricted by the voltage rating of individual power switches, which need to block the full dc link voltage. However, a series connection of power switches can be implemented to enable the 2L-VSI

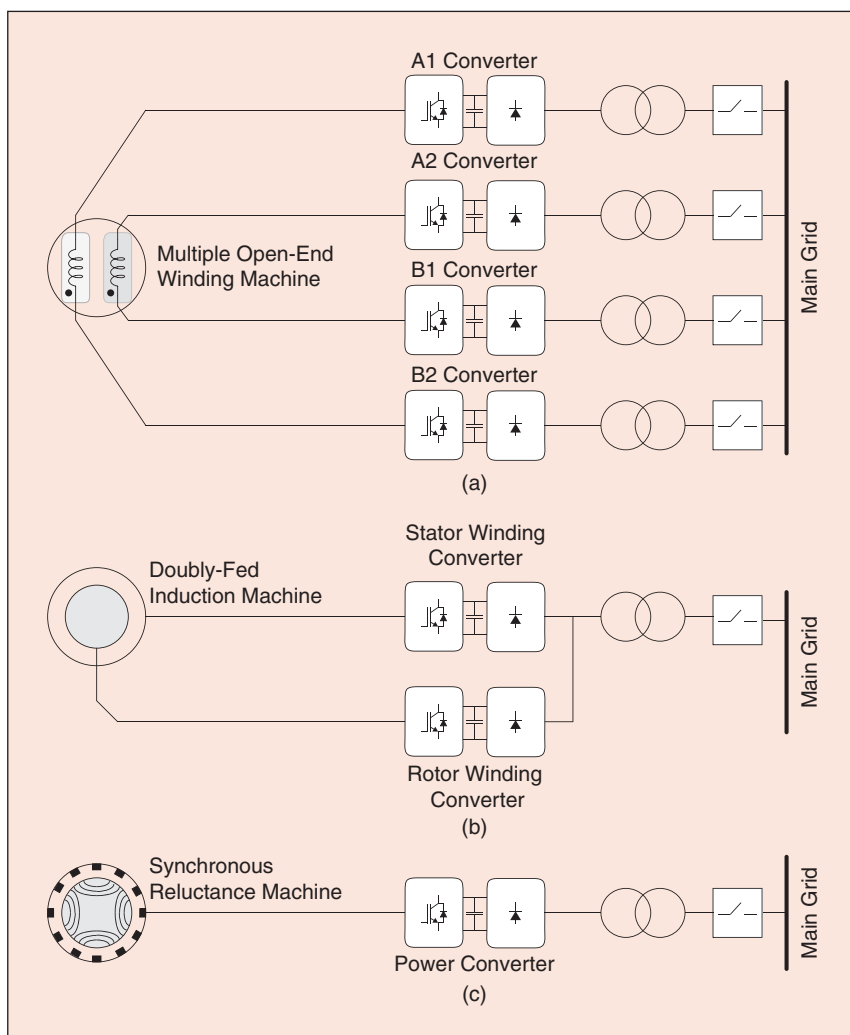


FIGURE 9 – Machine drive configurations. (a) The multiple open-end winding machine. (b) The doubly fed induction machine. (c) The synchronous reluctance machine.

topology to operate at higher voltage and power levels [42], [49].

Neutral-Point Clamped and Neutral-Point Piloted Multilevel Inverters

To overcome individual switch voltage limitations, multilevel topologies are the most common choice in high-power drive applications. In particular, the neutral-point clamped (NPC) architecture employs clamping diodes to connect the output to multiple voltage levels obtained by the cascaded arrange of dc capacitors, as detailed in Figure 10(b), for a three-level, three-phase NPC. More than three levels are also possible with a more complex topology, but the control complexity and cost of the NPC scale up very quickly with each increase in the number of output levels, hindering the application of this converter topology to higher power levels [50]. Transformerless converter topologies with a high power density, such as NPC, optimize the

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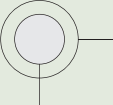
size and weight of the drive, which is particularly important to fit the machine and converter inside a vessel. The main producers of NPC converters are ABB [51], [52], Siemens [53], and Ingeteam [54], which provide voltages between 2.3 and 6.6 kV and power from 315 kW to 31.5 MW. The neutral-point piloted (NPP) inverter topology is able to increase the power density and power rating of the NPC, using the same IGBT power switches and control systems, and it is also capable of operating with lower switching losses than the NPC inverter [55], [56]. The structure of a three-level NPP is given in Figure 10(c), featuring

neutral-point connections through bidirectional switches, implemented with a series connection of IGBTs.

Cascaded H-Bridge

Modular topologies have become a widely accepted solution for high-power motor drives. Figure 11(a) depicts the structure of a generic cascaded H-bridge (CHB) converter with three cells per phase, with each cell consisting of a diode rectifier, a capacitive dc link filter, and an H-bridge [41]. The main advantage of CHB converters relates to their medium-voltage capability, which is made possible by the cascaded connection of cells. However,

TABLE 1 –ELECTRIC PROPULSION MACHINE DRIVES FOR MARINE APPLICATIONS.

TYPE	BRUSHED dc	INDUCTION	WRSM	PMSM	OPEN-END MULTIWINDING	DFIM	SynRM
Symbol							
Advantages	Maximum torque at any load condition Easy to implement torque and speed control	Robust and simple construction Almost maintenance free Wide range of load operations	Adjustable power factor Larger air gap compared to induction machines, becoming more shock resistant	High Power Density Maximum torque at zero speed	Improved torque quality Multiple degrees of redundancy and assurance of power segmentation	Operation in sub- and supersynchronous regions Enables downsizing the converter	Does not require permanent magnets or separate excitation Simple rotor design
Disadvantages	Commutator and brushes add complexity Significant maintenance costs High-power machines require additional interpoles Commutator isolation	Low efficiency at partial load Variable slip, depending on the load Low power factor at partial load	Needs separate excitation winding Brushes and slip rings add complexity Significant maintenance costs	Thermally sensitive High cost of magnets	Additional power converters Possibility of inducing zero sequence currents Complex interwinding and insulation layout	Slip rings and brushes add complexity Low efficiency due to rotor currents	Power factor depends on rotor design Only a reluctance torque component, leading to larger frames in high-power applications
Marine application	Type 23 frigates. 209-class submarines	Icebreaker, Offshore & Marine (Norway) LNG carrier, BG Group (Korea)	Albion-class landing platform dock Carnival Cruise Lines Mistral-class helicopter landing dock	212-class submarines Scorpene-class submarines	Type 45 destroyers <i>Queen Elizabeth</i> Invincible-class carrier	Type 26 frigates (under construction)	

LNG: liquified natural gas.

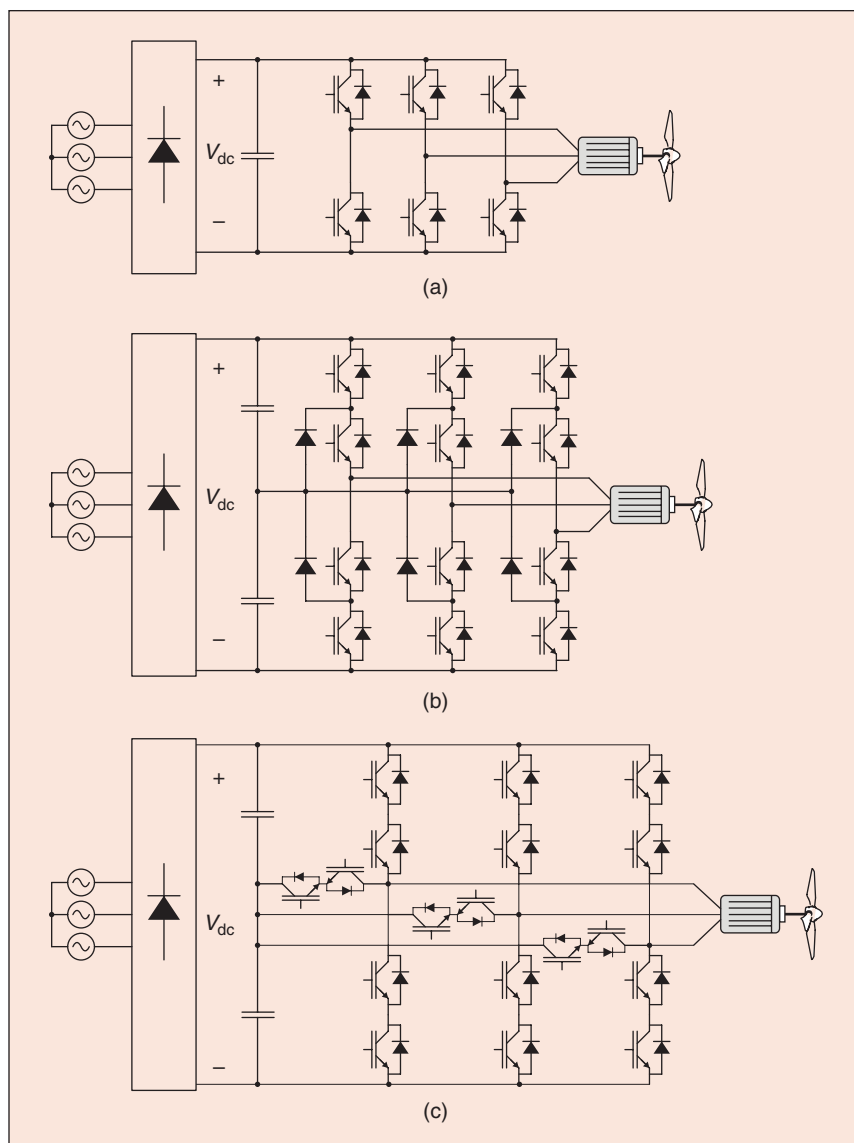


FIGURE 10 – VSIs. (a) Two level. (b) Neutral-point clamped. (c) Neutral-point piloted.

CHB drives are limited by the number of isolated dc power supplies required for each cell, which are implemented using bulky and expensive multiwinding transformers. The main provider of this type of converters is Siemens [57].

Modular Multilevel Converter

This converter is composed of several series-connected submodules grouped in six arms that are arranged symmetrically, as shown in Figure 11(b). The main advantage of the modular multilevel converter (MMC) stems from the high quality of the voltage and the currents at the input and output terminals and the ability to reach arbitrarily high voltage ratings by adding more submodules [58], [59]. Moreover, due to its inherent modularity and ability to operate under fault-tolerant strategies, the MMC is a promising technology for electric ship propulsion systems [60]. The main provider of this type of converters is Siemens [61].

CSIs

The main advantage of the CSI over the VSI relates to the CSI's medium-voltage capability, which is achieved by using simple two-level topologies, given the high voltage ratings of modern symmetric gate-commutated thyristors. A capacitive filter is required on the motor side of the inverter, as shown in Figure 12(a), to enable the forced switching of currents in inductive

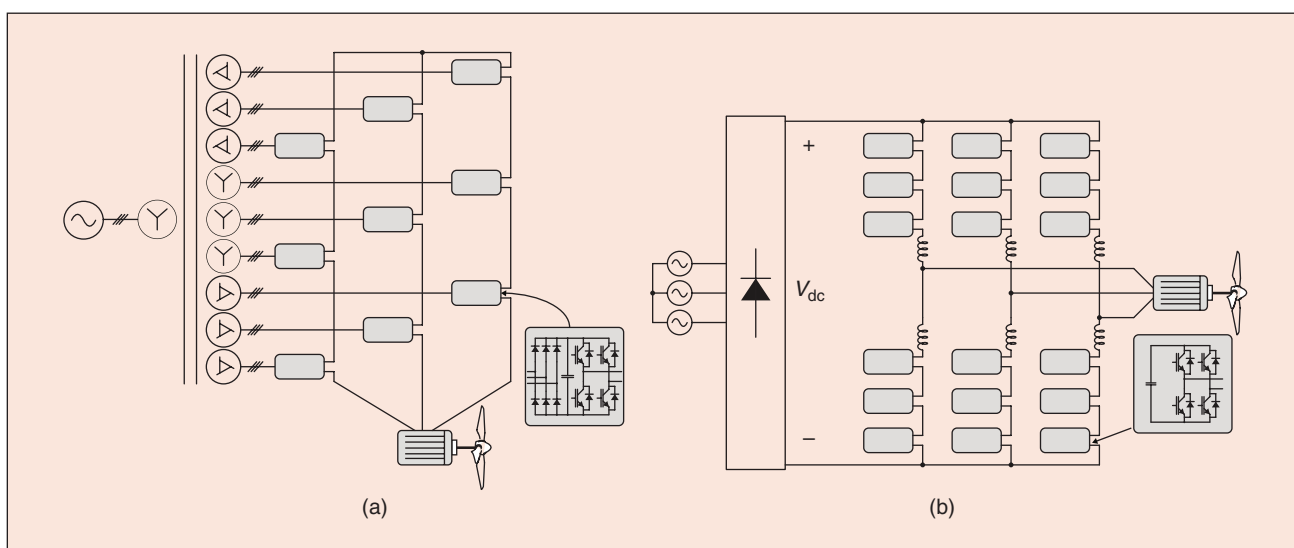


FIGURE 11 – Multicell converters. (a) The cascaded H-bridge converter. (b) The modular multilevel converter.

loads generating quasi-sinusoidal output waveforms. The output voltage waveforms with a reduced reduced voltage rate of change (dv/dt) are benign for ac motors, making possible the use of standard machines without enhanced insulation. The drawbacks of the CSI topology include a limited control bandwidth and a heavy weight due to bulky dc inductors. Moreover, to suppress common-mode voltage stress on the ac motor, an isolation transformer or common-mode voltage mitigation chokes must be employed [50]. The main supplier of CSIs for marine applications is Allen–Bradley [62].

Cycloconverter

Cycloconverters offer a high power capability due to the use of high-voltage, high-power SCRs, as documented in Figure 12(b). Their reliability is very high because thyristors are robust switching devices. However, these devices operate at a fundamental frequency, requiring large harmonics filters at both sides and reducing the power density and dynamical performance [63]. The main suppliers of this converter are ABB [64], Siemens [65], and General Electric [66], and they offer voltages up to 10 kV and power up to 150 MW.

Future Challenges

As a result of the harmful environmental impact of fossil fuels, more strict pollution policies, and the availability of high-performance machines and power electronics drives, the electric

Fuel cells are an emerging technology that, in recent years, has been evaluated for its use in marine applications.

propulsion of ships has become an exciting option for the transportation industry. However, to increase the use of these drives, challenges related to electrical propulsion technologies are discussed in this section.

Solar-Powered Ship Propulsion

Solar-powered propulsion systems can reduce the carbon footprint and reduce the cost of energy. Still, they are limited to the available surface required by the solar cells and by weather conditions. Therefore, solar-powered propulsion systems are currently being developed mostly for recreational marine craft, small tourist boats, and patrol vessels. Future projects based on solar-powered ships are the *Eoseas* concept cruise ship (builder: STH Europe), a zero-emission cruise vessel (builder: Deltamarine), the super eco-ship design (builder: Nippon Yusen Kabushiki Kaisha), and the Aquarius Eco Ship (builder: Eco Marine Power).

Fuel Cell Applications

Fuel cells are an emerging technology that, in recent years, has been evaluated for its use in marine applications [72]. They offer a variety of advantages over traditional power systems. Fuel

cells are efficient, environmentally friendly, and more reliable when compared to diesel and gas marine engines. However, in many cases, their cost is still too high for them to become a real alternative in marine propulsion applications. In this regard, several considerations have to be taken into account. Fuel availability is an essential factor for the application of fuel cells on ships. For submarines, pure hydrogen is a good choice because it can be obtained directly from the electrolysis of water. For surface ships, traditional fuels based on hydrocarbons have to be considered the first choice; hydrogen, on the other hand, introduces difficulties, such as storage and fueling operations in harbors.

Another aspect is referred to as the *propulsion power requirement*. For submarines, the total propulsion power requirement is less than 500 kW, becoming an attractive solution in terms of power density. However, for ships, power requirements are as high as 100 MW, introducing additional space and weight problems due to the required number of fuel cell arrangements [73], [74]. Fuel cells have been successfully implemented in 212- and 214-class submarines, using nine 34-kW proton-exchange

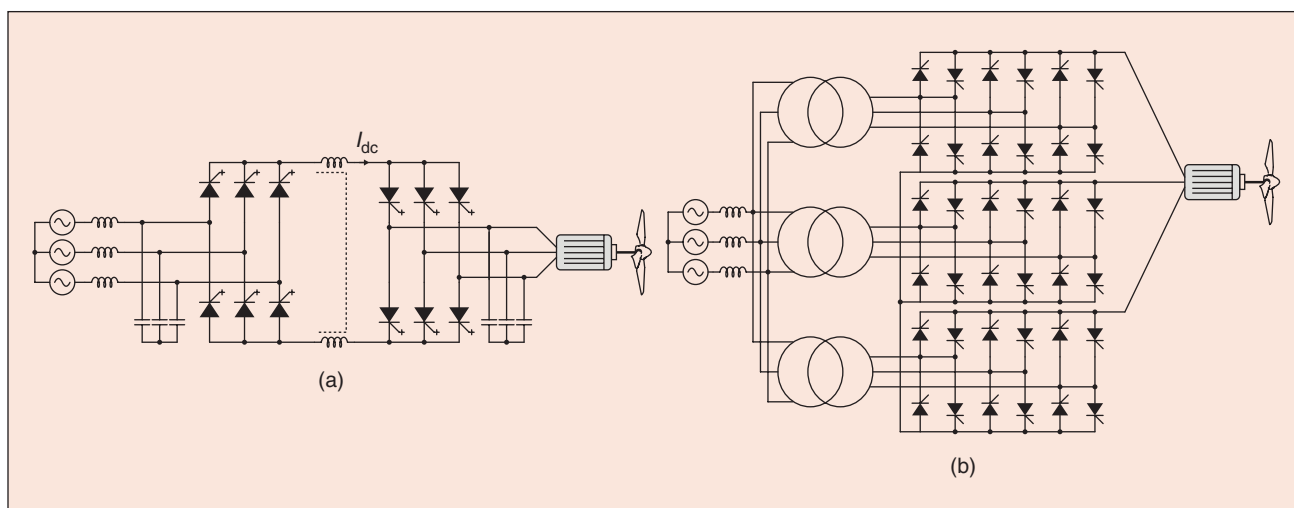


FIGURE 12 – Converter drive alternatives. (a) Current source. (b) The cycloconverter.

Electric marine propulsion systems provide several advantages compared to traditional gas turbines and diesel engines.

membrane (PEM) Siemens fuel cell modules (BZM34) in the first case, and two 120-kW PEM (BZM 120) Siemens fuel cell modules in the second case. Fuel cells for high-power systems on ships are still under development. Other limitations in the development of fuel cell-based electric propulsion systems in submarines and ships are primarily related to the availability of construction materials, the requirement for low-density and massive storage systems, and the need to increase fuel cells' life span.

DC Grid Integration

During the past five years, dc grid integration for onboard applications has attracted significant interest from both researchers and the shipbuilding industry because of the advantages it introduces, such as improved efficiency, the easy integration of different power sources, the reduced size and rating of switchboards, an absence of reactive power flows, and a system reconfiguration capability [75]. Global ship efficiency and emissions are improved since vessels' ac power generation units (diesel engines and turbogenerators) are decoupled from the dc distribution system and can be operated at an optimized speed, depending on the load demand. The platform support

vessel *Dina Star*, delivered in 2013 to the Myklebusthaug offshore platform, was the first ship powered by ABB that had a full dc grid system. However, dc applications have several technical issues, such as their circuit breaker technology, voltage stability, and power quality, which are difficult and challenging to resolve [76]–[78].

Energy Storage Integration

Electric energy storage employing batteries and hybrid battery–supercapacitor systems can improve the stability of IPSs and reduce voltage variations [79]. These systems can also be used to optimize fuel consumption to minimize costs and emissions [80], [81]. However, their main disadvantages concern their cost and power density [82] as well as their requirement for a high-performance control system to avoid overheating [23].

Conclusion

Electric marine propulsion systems provide several advantages compared to traditional gas turbines and diesel engines, such as a higher energy efficiency, reduced fuel consumption, and fewer harmful emissions. The necessary space for the installation of an electric propulsion system is smaller than that of a conventional

system, increasing the payload, providing more flexibility for machinery fitting, and enabling an independent design of the propeller and the drive. From the operational point of view, transmission stress and vibrations are reduced because there is no direct connection between the propeller and the prime mover; electric propulsion can provide high torque at zero speed and across the entire speed range, improving ship maneuverability. All these advantages come with additional challenges, including training ship crews and a higher level of automation, which, added to higher initial costs, has slowed the adoption of this new technology.

A large variety of machines, converter topologies, and drive configurations have been examined in this article (see Table 2). That equipment has become a critical technology for the integration of more electric ship propulsion systems into the maritime industry. Renewable energy resources that can increase vessels' efficiency and reduce emissions and massive energy storage systems are two technologies that can extend the advantages of electric propulsion systems, but they require further research.

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TABLE 2 – EXAMPLE APPLICATIONS OF POWER CONVERTERS IN MARINE PROPULSION.

VESSEL NAME	OASIS OF THE SEAS [67]	WOODSIDE ROGERS [68]	JUAN CARLOS I [69]	SK ATOMIK [70]	MS VIKKI [71]
Vessel type	Cruise liner	LNG carrier	Aircraft carrier	Offshore supply	LNG-fueled bulk carrier
Propulsion	3 × 20 MW Azipod thruster	2 × 13.26 MW	2 × 11 MW		1.25 MW Auxiliary power drive
Converter model	ACS 6000	MV 7000	GM150	Lean drive	WE drive
Manufacturer	ABB	General Electric	Siemens	STADT	Yaskawa/The Switch, WE Tech
Rated power	5–36 MW	3–81 MW	17–23 MW	0.1–30 MW	0.5–6 MW
Voltage	2.3–3.3 kV	Up to 13.8 kV	3.3–6.6 kV	Up to 15 kV	400–690 V
Frequency	0–75 Hz	15–90 Hz	0–70 Hz		
Configuration	Three-level NPC (IGCT)	Three-level NPC (IGBT)	Five-level NPC (IGCT)		AFE/VSF

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